

HAOJIE HUANG

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EDUCATION

Doctor of Philosophy | *Computer Science*
Northeastern University

Sep. 2021 – Expected Dec. 2026
Boston, MA

Master of Science | *Robotics*
Northeastern University

Sep. 2018 – May 2021
Boston, MA

Bachelor of Engineering | *Mechanical Engineering*
Shandong University

Sep. 2013 – May 2017
Jinan, China

RESEARCH AREA

Robot Learning, Geometric Deep Learning, Manipulation, Symmetry / Equivariance, Action Representation, 2/3D Perception, Reinforcement Learning, Vision Language and Action Learning

RESEARCH HIGHLIGHTS

Leading researcher in **Symmetric Manipulation Learning** and **Geometric Action Representation**, where

- the first area enables fast and efficient learning
- the second area transfers the learning domain to a simple and natural space

Pioneered several **original approaches** to manipulation policy learning:

- First to systematically realize symmetries for manipulation learning
- First to propose a 3D generative world action model for manipulation learning

WORK EXPERIENCE

Applied Scientist Intern
Amazon Robotics

Sep 2024 – Jan 2025
North Reading, MA

- Worked on closed-loop Imitation learning algorithm

Research Scientist Intern
Boston Dynamics AI Institute

Sep 2023 – Jan 2024
Cambridge, MA

- Worked on the machine intelligence and robot learning algorithm

R&D Intern
SharkNinja Corp.

June 2019 – Dec 2019
Needham, MA

- Worked on the hardware(sensors, motors and etc.) and software(control algorithms, path planning and etc.) for robotic vacuums

TEACHING EXPERIENCE

Robotic Science and System (CS5335-Teaching Assistant)
Northeastern University

Spring 2021
Boston, MA

AWARD

Best Paper Award - Honorable Mention on IEEE Robotics and Automation Letters

Vienna, Austria

June 2026

Outstanding Paper Finalist at Conference on Robot Learning (CoRL) 2024

Munich, Germany

Nov 2024

Best Paper Award Finalist on Scaling Robot Learning Workshop of ICRA 2022

Philadelphia, United States

May 2022

PROFESSIONAL CONTRIBUTION

2027 Associate Editor: ICRA; **Reviewer:** ICLR.

2026 Reviewer: ICLR, ICML, NeurIPS, ICRA, RSS, CoRL, RAL, TRO. **Invited Talks:** Tsinghua University.

2025 Reviewer: ICLR, ICML, NeurIPS, ICRA, IROS, RSS, CoRL, RAL, TRO. **Invited Talks:** Amazon Robotics. **Paper Presentations:** ICRA 2025.

2024 Reviewer: ICRA, IROS, RAL. **Invited Talks:** Worcester Polytechnic Institute (WPI); Next-Gen Robot Learning Symposium, TU Darmstadt. **Paper Presentations:** ICLR 2024, CoRL 2024.

2023 Reviewer: ICRA, IROS, RAL. **Paper Presentations:** ICRA 2023. **Program Committee:** Workshop on Symmetries in Robot Learning, RSS 2023.

2022 Reviewer: ICRA, IROS, RAL. **Paper Presentations:** ICRA 2022 (live demo and talk); RSS 2022; RLDM 2022.

FIRST-AUTHOR PUBLICATIONS

Action Map Policy: Learning 3D Closed-loop Manipulation via Pixel Classification

Jul 2026

Haojie Huang et al. (*Contribution: first author*)

- Arxiv Preprint -Under Submission

Pix2Act: Image-Space Manipulation Policies with Equivariant Augmentation

Jul 2026

Haojie Huang et al. (*Contribution: first author*)

- Arxiv Preprint -Under Submission

Match Policy: A Simple Pipeline from Point Cloud Registration to Manipulation Policies

May 2025

Haojie Huang et al. (*Contribution: first author*)

- International Conference on Robotics and Automation, Atlanta, ICRA 2025

Learning efficient and robust language-conditioned manipulation using textual-visual relevancy and equivariant language mapping

June 2024

Mingxi Jia et al. (*Contribution: co-first author*)

- IEEE Robotics and Automation Letters 2025
- Best Paper Award Honorable Mention of IEEE Robotics and Automation Letters (5 out of 1700 publications)

Imagination Policy: Using Generative Point Cloud Models for Manipulation Policies

June 2024

Haojie Huang et al. (*Contribution: first author*)

- Conference on Robot Learning, Munich, CoRL 2024

Fourier Transporter: Bi-Equivariant Robotic Manipulation in 3D Haojie Huang et al. <i>(Contribution: first author)</i>	Sep 2023
• International Conference on Learning Representations, Vienna, ICLR 2024	
Language-conditioned Equivariant Grasp Haojie Huang et al. <i>(Contribution: first author)</i>	Sep 2023
• Arxiv Preprint	
Leveraging Symmetries in Pick and Place Haojie Huang et al. <i>(Contribution: first author)</i>	March 2023
• International Journal of Robotics Research, Volume 43, Issue 4, 2024	
Edge Grasp Network: A Graph-Based SE(3)-invariant Approach to Grasp Detection Haojie Huang et al. <i>(Contribution: first author)</i>	Dec 2022
• International Conference on Robotics and Automation, London, ICRA 2023	
• Sim-to-Real Robot Learning, CoRL 22 workshop, Auckland	
• Symmetries in Robot Learning, RSS 2023 workshop, Daegu	
Equivariant Transporter Network Haojie Huang et al. <i>(Contribution: first author)</i>	June 2022
• Robotics Science and System, New York, RSS 2022	
• Scaling Robot Learning, ICRA 22 workshop, Philadelphia, Best Paper Award Finalists	
• Reinforcement Learning and Decision Making, RLDM 2022, Rhode Island	
Graph Attention Shape Completion Network Haojie Huang et al. <i>(Contribution: first author)</i>	Dec 2021
• International Conference on 3D Vision, London, 3DV 2021	

COLLABORATED PUBLICATIONS

Retriever: Composing Closed-Loop Asynchronous Robot Programs Linfeng Zhao et al. <i>(Contribution: collaborator)</i>	Jul 2026
• Arxiv Preprint - under review	
Scale Up Strategically: Learning Compositional Generalization via Bias-Aware Evaluation and Data Collection for Robotic Manipulation Yu Qi et al. <i>(Contribution: collaborator)</i>	Jul 2026
• Conference on Robot Learning, Austin, CoRL 2026	
Evaluating Reasoning Coherence in Video Generative Models with Text and Visual Hints Yu Qi et al. <i>(Contribution: collaborator)</i>	March 2026
• European Conference on Computer Vision - ECCV 2026	
Equivariant diffusion policy for sample-efficient robotic manipulation Dian Wang et al. <i>(Contribution: collaborator)</i>	Feb 2026
• International Journal of Robotics Research, IJRR 2026	
Generalizable Hierarchical Skill Learning via Object-Centric Representation Haibo Zhao et al. <i>(Contribution: collaborator)</i>	Oct 2025
• IEEE Robotics and Automation Letters 2026	

Accelerating Model-Based Reinforcement Learning Using Equivariance Ahmed Agha et al. <u>(Contribution: collaborator)</u> • Northeast Robotics Colloquium - NERC 2025	Sep 2025
Clebsch-Gordan Transformer: Fast and Global Equivariant Attention Owen Howell et al. <u>(Contribution: collaborator)</u> • Arxiv Preprint	Sep 2025
BEAR: Benchmarking and Enhancing Multimodal Language Models for Embodied AI Yu Qi et al. <u>(Contribution: senior collaborator)</u> • International Conference on Machine Learning - ICML 2026	Sep 2025
Equivariant Goal Conditioned Contrastive Reinforcement Learning Arsh Tangri et al. <u>(Contribution: senior collaborator)</u> • Arxiv Preprint	July 2025
3D Equivariant Visuomotor Policy Learning via Spherical Projection Boce Hu et al. <u>(Contribution: collaborator)</u> • Conference on Neural Information Processing Systems, San Diego , NeurIPS 2025 • NeurIPS Spotlight - One of 688 Spotlight papers among 21,575 main-track submissions	June 2025
Push-Grasp Policy Learning Using Equivariant Models and Grasp Score Optimization Boce Hu et al. <u>(Contribution: collaborator)</u> • IEEE Robotics and Automation Letters 2025 - RAL 25	April 2025
Coarse-to-Fine 3D Keyframe Transporter Xupeng Zhu et al. <u>(Contribution: collaborator)</u> • Arxiv preprint	Feb 2025
OrbitGrasp: SE(3)-Equivariant Grasp Learning Boce Hu et al. <u>(Contribution: co-second author)</u> • Conference on Robot Learning, Munich, CoRL 2024	June 2024
Equivariant Diffusion Policy Dian Wang et al. <u>(Contribution: collaborator)</u> • Conference on Robot Learning, Munich, CoRL 2024, Outstanding Paper Finalist	June 2024
ThinkGrasp: A Vision-Language System for Strategic Part Grasping in Clutter Yaoyao Qian et al. <u>(Contribution: collaborator)</u> • Conference on Robot Learning, Munich, CoRL 2024	June 2024